

CHE 201 Spring 2020 HW 6

Anna Christensen

TOTAL POINTS

99 / 100

QUESTION 1

1 Question 1 14 / 15

- 0 pts Correct
- ✓ - 1 pts Wrong unit
- 9 pts Wrong volume of air per kilogram
- 6 pts Wrong understanding of volumetric flow rate

- 5 pts Not enough evaluation on economic factors
- 2.5 pts Not enough evaluation on the effects on the ecosystem
- 5 pts Not enough evaluation on the effects on the ecosystem
- 15 pts Not answered

QUESTION 2

2 Question 2 15 / 15

- ✓ - 0 pts Correct

QUESTION 3

3 Question 3 15 / 15

- ✓ - 0 pts Correct
- 2.5 pts a) Wrong calculation
- 2.5 pts a) Wrong equation
- 2.5 pts b) Wrong calculation
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- 2.5 pts c) Wrong calculation
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QUESTION 4

4 Question 4 20 / 20

- ✓ - 0 pts Correct

QUESTION 5

5 Question 5 20 / 20

- ✓ - 0 pts Correct

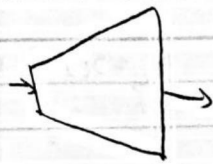
QUESTION 6

6 Question 6 15 / 15

- ✓ - 0 pts Correct
- 2.5 pts Not enough evaluation on technical factors
- 5 pts Not enough evaluation on technical factors
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HW 6 3/10/2020

4.4 ①



$$AV = \dot{m}v \rightarrow \frac{\text{m}^3}{\text{kg}}$$

$$PV = RT \rightarrow v = \frac{RT}{P}$$

$$P_2 = 200 \text{ kPa}$$

$$T_2 = 150^\circ \text{C}$$

$$\dot{m}_2 = 11.5 \text{ kg/s}$$

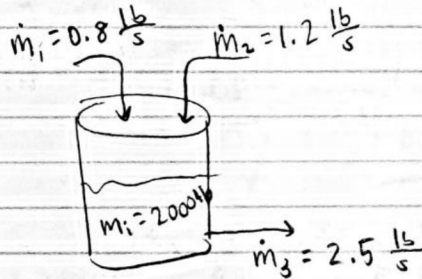
$$AV = ?$$

$$R = 287 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$v = \frac{(287 \frac{\text{J}}{\text{kg} \cdot \text{K}})(150 + 273 \text{ K})}{200 \times 10^3 \text{ Pa}} = 0.6070 \frac{\text{m}^3}{\text{kg}}$$

$$AV = (11.5 \frac{\text{kg}}{\text{s}})(0.607 \frac{\text{m}^3}{\text{kg}}) = \boxed{6.98}$$

4.6 ②



$$\frac{dm_{cv}}{dt} = \dot{m}_1 + \dot{m}_2 - \dot{m}_3$$

$$\int_0^t dm_{cv} = \int_0^t (\dot{m}_1 + \dot{m}_2 - \dot{m}_3) dt$$

$$m_{cv}(t) - m_{cv}(0) = (\dot{m}_1 + \dot{m}_2 - \dot{m}_3)t$$

$$m_{cv}(t) = m_{cv}(0) + (\dot{m}_1 + \dot{m}_2 - \dot{m}_3)t$$

$$m_{cv}(1 \text{ hr}) = 2000 \text{ lb} + (0.8 + 1.2 - 2.5)(1 \text{ hr} \times \frac{3600 \text{ s}}{1 \text{ hr}})$$

$$\boxed{m_{cv}(1 \text{ hr}) = 200 \text{ lb}}$$

1 Question 1 14 / 15

- 0 pts Correct

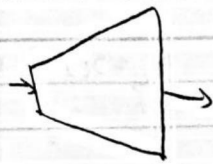
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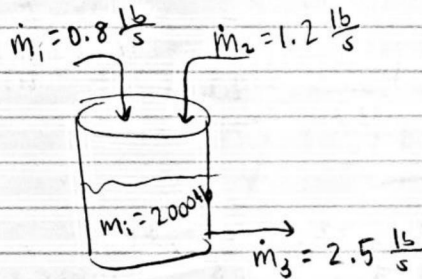
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$$m_{cv}(t) - m_{cv}(0) = (\dot{m}_1 + \dot{m}_2 - \dot{m}_3)t$$

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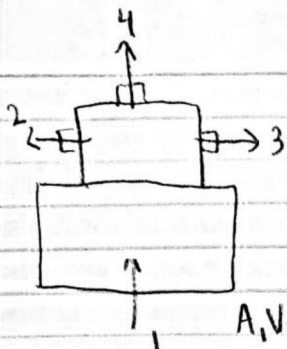
$$m_{cv}(1 \text{ hr}) = 2000 \text{ lb} + (0.8 + 1.2 - 2.5)(1 \text{ hr} \times \frac{3600 \text{ s}}{1 \text{ hr}})$$

$$\boxed{m_{cv}(1 \text{ hr}) = 200 \text{ lb}}$$

2 Question 2 15 / 15

✓ - 0 pts Correct

4.17 (3)



$$T_2 = T_3 = T_4 = 80^\circ\text{F}$$

$$D_2 = D_3 = 26\text{ in} \quad D_4 = 50\text{ in}$$

$$V_2 = V_3 = 10\text{ ft/s}$$

$$A_1 V_1 = 15000 \frac{\text{ft}^3}{\text{min}}$$

$$P_1 = P_2 = P_3 = P_4 = 1\text{ atm}$$

$$T_1 = 35^\circ\text{F} = 274.817\text{ K}$$

$$1\text{ atm} = 1.01325 \times 10^5 \frac{\text{N}}{\text{m}^2}$$

$$R = 1.31443 \frac{\text{atm} \cdot \text{ft}^3}{\text{K} \cdot \text{lb}}$$

* a) $PV = RT \rightarrow V = \frac{RT}{P} \rightarrow \frac{(1.31443)(274.817\text{ K})}{1545 \frac{\text{lb}}{16\text{ mol}} / 28.97 \frac{\text{lb}}{16\text{ mol}}} = 12.47 \frac{\text{ft}^3}{\text{lb}}$

$$m = \frac{AV}{V}$$

$$\dot{m} = \frac{15000 \frac{\text{ft}^3}{\text{min}}}{12.47 \frac{\text{ft}^3}{\text{lb}}} \times \frac{1\text{ min}}{60\text{ s}} = 20.05 \frac{\text{lb}}{\text{s}} = \dot{m}_1$$

b) $A_2 V_2 = A_3 V_3 = \pi \left(\frac{26\text{ in}}{2} \right)^2 \left(10 \frac{\text{ft}}{\text{s}} \times \frac{\text{ft}^2}{144\text{ in}^2} \times \frac{60\text{ s}}{\text{min}} \right)$

$$= 2212 \frac{\text{ft}^3}{\text{min}} = A_2 V_2 = A_3 V_3$$

c) $\dot{m}_1 = \dot{m}_2 + \dot{m}_3 + \dot{m}_4$

$$\dot{m}_4 = \dot{m}_1 - \dot{m}_2 - \dot{m}_3$$

$$\dot{m}_4 = 20.05 - (2 \times 2.71 \frac{\text{lb}}{\text{s}})$$

$$\dot{m}_4 = 14.63 \frac{\text{lb}}{\text{s}}$$

$$\dot{m}_2 = \dot{m}_3 = \frac{AVP}{RT}$$

$$= \left(2212 \frac{\text{ft}^3}{\text{min}} \right) \left(1\text{ atm} \times \frac{2116.24 \frac{\text{lb}}{\text{ft}^2}}{1\text{ atm}} \right)$$

$$\left(\frac{1545 \frac{\text{lb}}{16\text{ mol}}}{28.97 \frac{\text{lb}}{16\text{ mol}}} \right) \times (299.8\text{ K})$$

$$\times \frac{1\text{ min}}{60\text{ s}} = 2.71 \frac{\text{lb}}{\text{s}} = \dot{m}_2 = \dot{m}_3$$

$$V = \frac{RT}{P} = \frac{1545}{28.97} \times 299.8$$

$$\frac{1\text{ atm}}{2116.24 \frac{\text{lb}}{\text{ft}^2}} = 13.60 \frac{\text{ft}^3}{\text{lb}} = V$$

$$\dot{m} = \frac{AV}{v}$$

$$V = \frac{\dot{m} v}{A} = \frac{(14.63 \frac{\text{lb}}{\text{s}})(13.60 \frac{\text{ft}^3}{\text{lb}})}{\pi \left(\frac{50 \text{ in}}{2}\right)^2} \times \frac{144 \text{ in}^2}{\text{ft}^2} = \boxed{14.59 \frac{\text{ft}}{\text{s}}}$$

1 (4) $T_1 = 25^\circ\text{C}$

$P_1 = 100 \text{ kPa}$

$AV = 23 \frac{\text{m}^3}{\text{h}}$

$V = 120 \text{ V}$

$I = 4 \text{ A}$

$C_p = 1.005 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}((h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g(z_1 - z_2))$$

$$h_1 - h_2 = C_p(T_2 - T_1)$$

a) mass flow rate?

$$\dot{m} = \frac{AV}{v} = \frac{P_1(AV)}{RT_1}$$

$$= \frac{(100 \times 10^3 \text{ Pa})(23 \frac{\text{m}^3}{\text{hr}})}{287 \frac{\text{J}}{\text{kg} \cdot \text{K}} \times (25 + 273 \text{ K})} = \boxed{26.9 \frac{\text{kg}}{\text{hr}}}$$

b) $\dot{W}_{cv} = \dot{m}(h_1 - h_2)$

Power = $W = -VI$

$= -(120 \text{ V})(4 \text{ A})$

$= -480 \text{ W}$

$P = W = -0.48 \text{ kW}$

$= -0.48 \text{ kJ}$

$W = \dot{m} C_p(T_1 - T_2)$

$-0.48 = 26.9 \frac{\text{kg}}{\text{hr}} \times \frac{1 \text{ hr}}{3600 \text{ s}} \times 1.005 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$

$\times ((25 + 273) - T_2)$

$T_2 = 361.9 \text{ K} = \boxed{88.9^\circ\text{C} \approx T_2}$

3 Question 3 15 / 15

✓ - 0 pts Correct

- 2.5 pts a) Wrong calculation
- 2.5 pts a) Wrong equation
- 2.5 pts b) Wrong calculation
- 2.5 pts b) Wrong equation
- 2.5 pts c) Wrong calculation
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$$\dot{m} = \frac{AV}{v}$$

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a) mass flow rate?

$$\dot{m} = \frac{AV}{v} = \frac{P_1(AV)}{RT_1}$$

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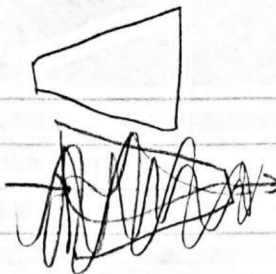
$\times ((25 + 273) - T_2)$

$T_2 = 361.9 \text{ K} = 88.9^\circ\text{C} = T_2$

4 Question 4 20 / 20

✓ - 0 pts Correct

9 ⑤ $P_1 = 200 \text{ kPa}$
 $T_1 = 325 \text{ K}$
 $\dot{m}_1 = 0.5 \frac{\text{kg}}{\text{s}} = \dot{m}_2$
 $A_1 = 6 \text{ cm}^2$



$P_2 = 100 \text{ kPa}$
 $V_2 = 250 \frac{\text{m}}{\text{s}}$
 $C_p = 1.008 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$

a) $\dot{m} = \frac{A_1 V_1}{V} = \frac{A_1 V_1 P_1}{R T_1}$ $PV = nRT$
 $\frac{L}{V} = \frac{P}{nRT}$
 $0.5 \frac{\text{kg}}{\text{s}} = \frac{(6 \times 10^{-4} \text{ m}^2) (V_1) (200 \times 10^3 \text{ Pa})}{(8.314 \frac{\text{m}^3 \text{ Pa}}{\text{K} \cdot \text{mol}} \times \frac{1 \text{ mol}}{0.0289 \text{ kg}}) (325)}$

$V_1 = 389 \frac{\text{m}}{\text{s}}$

b) $0 = \cancel{\dot{Q}_{cv}} - \cancel{\dot{W}_{cv}} + \dot{m}[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g(\cancel{z_1} - \cancel{z_2})]$

$\dot{m}(h_1 - h_2) = - \frac{V_1^2 - V_2^2}{2}$

$h_1 - h_2 = C_p(T_1 - T_2) = \frac{V_2^2 - V_1^2}{2 \times 1 \text{ kg/s}}$

$1.008 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} (325 - T_2) = \frac{250^2 - 389^2}{2000}$

$T_2 = 369 \text{ K}$

c) $\dot{m}_2 = \frac{A_2 V_2}{V} = \frac{A_2 V_2 P}{R T} = \frac{(100 \times 10^3) (250 \frac{\text{m}}{\text{s}}) (A_2)}{(8.314 \times \frac{1 \text{ mol}}{0.0289 \text{ kg}}) (369 \text{ K})} = 0.5 \frac{\text{kg}}{\text{s}}$

$A_2 = 21.2 \text{ cm}^2$

5 Question 5 20 / 20

✓ - 0 pts Correct

6)

It is estimated that the cost of wave energy between 9 and 16 cents per kilowatt-hour. For tidal energy, the cost is lower, between 6 to 9 cents. Neither of these are as cost effective as coal, which is 3 cents per kilowatt-hour. For environmental issues, the placing of a man-made structure could have an effect on marine life.

Tidal barrages, which capture the energy from tidal forces of water moving in and out of a bay or river, are the largest challenges. These turbine structures could harm the fish and impede their migration. They also interfere with the normal flushing of silt and other dissolved pollutants, which would affect water quality, hurting the wildlife.

For wave energy, the equipment is not survivable for a long time and may release lubricants into the water. For ocean thermal energy, the cost is very expensive and is limited to certain areas in tropical oceans and we must be able to transmit electricity there.

<https://science.howstuffworks.com/environmental/green-tech/energy-production/ocean-power4.htm>

The complexity of harnessing wave power has been a challenge. Some designs generate electricity on the spot and transmit it through undersea cables and others pass the mechanical energy of the wave along to land before turning it into electrical energy. It is not yet clear which method is more efficient. Operating in the ocean is far more difficult than operation on land, as building installations offshore is more expensive than building on shore. Also, saltwater is a hostile environment for devices, and the waves are difficult to work with since they do not always go in one direction. Waves not only roll past the device, they bob up and down and can converge on all sides, which is a challenge for the equipment design. The fundamental roadblock for larger scale usage is the cost.

https://e360.yale.edu/features/why_wave_power_has_lagged_far_behind_as_energy_source

Another obstacle with wave resource study is the lack of long term wave measurements with different waves. This makes it difficult to specify the optimum locations for wave energy. An ideal spot would supply consistent power throughout the year, which would be decided by seasonal weather and wind patterns. All of these factors can vary, however, and waves below or above what the equipment is designed for would reduce the wave energy output. To overload the operation could also lead to equipment damage. These factors are unavoidable.

The lack of coordinated efforts in advancing technology also inhibits the advancing of this new form of energy conversion. Wave energy is still not widely recognized in North America as an alternative source of renewable energy.

<https://www.oceanenergycouncil.com/challenges-issues-wave-energy-conversion/>

6 Question 6 15 / 15

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